

Updated Data Dictionary

Global Superstore Modelling Dataset — Week 3
AnalystLab Africa Data Science Internship | Analyst: Ozoeze Wilord Ugonna

1. Dataset Overview

This data dictionary describes the **final modelling dataset** produced in Week 3: **global_superstore_modelling_dataset.csv**. It contains selected original fields from the Global Superstore source data plus engineered features created for advanced analysis and Week 4 machine learning.

Item	Details
File name	global_superstore_modelling_dataset.csv
Rows	51,290 orders
Columns	21 fields (original + engineered)
Primary business outcome	Profit / Is_Loss_Making
Source	Global Superstore dataset (cleaned in Week 2; refined in Week 3)

2. Field Dictionary

Columns are grouped as **Original** (from source data) or **Engineered** (created in Week 3). Data types reflect the modelling file after preparation.

Field Name	Type	Source	Description
Order Date	datetime	Original	Date the order was placed.
Segment	categorical	Original	Customer segment: Consumer, Corporate, or Home Office.
City	categorical	Original	City where the order was delivered.
State	categorical	Original	State or province of the delivery location.
Country	categorical	Original	Country of the order.
Market	categorical	Original	High-level market region (e.g. APAC, EU, US, LATAM, EMEA, Africa, Canada).
Region	categorical	Original	Sub-region within the market.
Category	categorical	Original	Product category: Furniture, Office Supplies, or Technology.
Sub-Category	categorical	Original	Finer product grouping within Category (e.g. Phones, Chairs, Binders).
Sales	numeric	Original	Sales revenue for the order line (currency units as in source data).
Quantity	integer	Original	Number of units ordered.
Discount	numeric	Original	Discount rate applied to the order (0 to 1 scale, e.g. 0.2 = 20%).
Profit	numeric	Original	Profit (or loss) for the order line. Primary continuous target variable.
Shipping Cost	numeric	Original	Cost of shipping the order.
Ship Mode	categorical	Original	Shipping method: Standard Class, Second Class, First Class, or Same Day.
Order Priority	categorical	Original	Order priority level: Low, Medium, High, or Critical.
Profit_Margin	numeric	Engineered	Profit ÷ Sales. Profitability rate; comparable across order sizes. Do not use as input if predicting Profit.
Is_Loss_Making	binary (0/1)	Engineered	1 if Profit < 0, else 0. Loss flag and potential classification target (~24.5% positive).
Discount_Impact	numeric	Engineered	Sales × Discount. Estimated absolute revenue given away through discounting.
Shipping_Efficiency	numeric	Engineered	Shipping Cost ÷ Sales. Relative shipping burden on order value.
Days_to_Ship	integer	Engineered	Ship Date – Order Date in days. Fulfilment speed / operational lag.

3. Target Variables and Leakage Rules

- **Profit** — continuous outcome for regression models.
- **Is_Loss_Making** — binary outcome for classification models.
- If predicting **Is_Loss_Making**, do **not** use Profit or Profit_Margin as model inputs (target leakage).
- If predicting **Profit**, do **not** use Profit_Margin or Is_Loss_Making as model inputs.

4. Fields Excluded from Modelling Dataset

The following source fields were intentionally excluded from the modelling file, with justification:

Excluded field	Reason
Row ID, Order ID	Pure identifiers; no predictive value.
Customer ID, Customer Name	High cardinality; not generalisable; privacy.
Product ID, Product Name	Very high cardinality; unstable for modelling.
Postal Code	Missing for ~80% of rows (mostly non-US).
Ship Date (raw)	Encoded as Days_to_Ship; raw date retained only via Order Date where needed.

5. Version and Usage Notes

- **Version:** Week 3 final modelling dataset (post feature engineering and selection).
- **Upstream:** Built from the Week 2 cleaned Global Superstore extract; full Week 2 cleaning was not repeated.
- **Intended use:** Exploratory analysis, statistical validation follow-up, and Week 4 supervised learning.
- **Known limitation:** Geographic granularity outside the US is limited by missing Postal Code.

This data dictionary supports reproducibility and clear communication of the modelling dataset for AnalystLab Africa Week 3 submission and Week 4 model development.